

Ridge Regression: The Maths and Code from Scratch

Contents

1. Overview	2
2. One feature	3
2.1 The loss	3
2.2 The intercept	3
2.3 The slope	4
2.4 Why the slope shrinks	5
3. Many features	6
3.1 The loss in matrix form	7
3.2 Expanding and differentiating	7
3.3 Solving for w	7
3.4 Not penalising the intercept	8
4. Summary	11
5. Sources	11
6. Key terms	11

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Standardization ([Note ML-009](#)); Multiple linear regression ([Note ML-052](#)); Normal equation ([Note ML-053](#)); Gradient descent ([Note ML-056](#)); Bias-variance trade-off ([Note ML-061](#)); Regularisation ([Note ML-062](#)).
- **Leads to:** Elastic Net ([Note ML-068](#)).
- **Compare with:** Lasso regression ([Note ML-066](#)); L1 and L2 regularisation in neural networks ([Note DL-026](#)).

1. Overview

Key point: Adding the Ridge penalty to the loss and setting the derivative to zero gives a closed-form answer. With one feature the slope becomes $m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2 + \lambda}$; with many features it becomes $\beta = (X^T X + \lambda I)^{-1} X^T y$.

The previous Note explained what Ridge does: it adds λ times the squared coefficients to the loss, which keeps them small. This Note derives the formulas that find the Ridge coefficients, the same way the OLS and normal equation Notes did for plain linear regression.

Here a **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) y is the value we predict. There are two cases:

- **One feature:** a formula for the slope m and the intercept b .
- **Many features:** a matrix formula for all coefficients at once.

Both are then coded from scratch and checked against scikit-learn's **Ridge**.

Figure 1 shows what the formulas will produce on the 100-observation example of the previous Note. The ordinary least-squares line has slope 27.83. With $\lambda = 10$ the **Ridge regression** (G-1691) line tilts down to 24.95, and with $\lambda = 100$ to 12.93. All three lines pass through the same point, the point of means; Section 2 shows why.

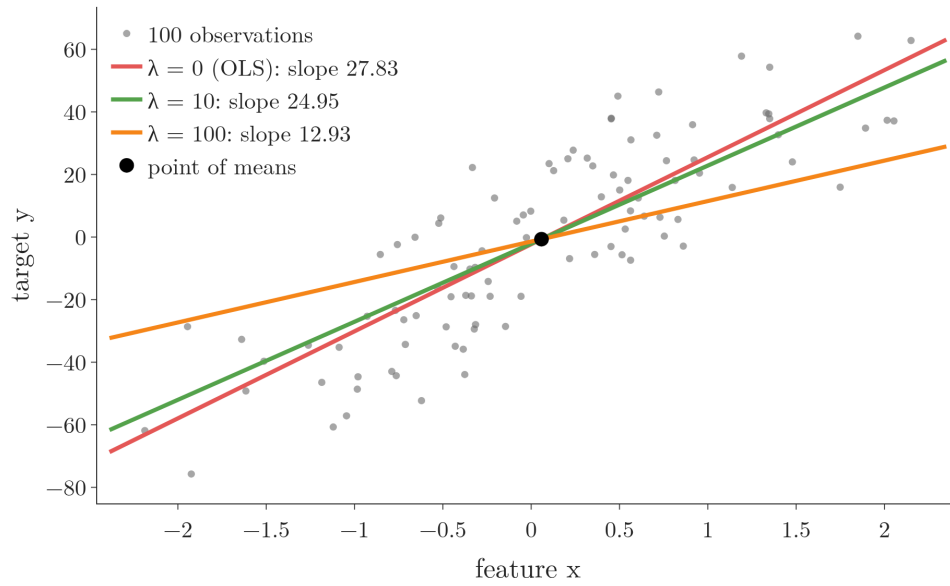


Figure 1: Ridge lines on the 100-observation example: $\lambda = 0$ (ordinary least squares, slope 27.83), $\lambda = 10$ (24.95) and $\lambda = 100$ (12.93). Every line passes through the point of means.

2. One feature

Key point: The Ridge slope is the OLS slope with λ added to the bottom of the fraction. The intercept formula does not change.

2.1 The loss

Key point: The loss is the usual sum of squared errors plus λm^2 .

With one feature, the prediction is $\hat{y}_i = mx_i + b$. The Ridge loss is

$$L = \sum_{i=1}^n (y_i - mx_i - b)^2 + \lambda m^2$$

Only the slope m is penalised, not the intercept b .

2.2 The intercept

Key point: The penalty does not contain b , so the derivative with respect to b is the same as in OLS, and so is the answer.

Differentiate L with respect to b and set it to zero. The term λm^2 does not contain b , so it disappears:

$$\frac{\partial L}{\partial b} = -2 \sum_{i=1}^n (y_i - mx_i - b) = 0$$

This derivative is exactly the OLS step, and it gives the same result:

$$b = \bar{y} - m\bar{x}$$

2.3 The slope

Key point: Differentiating with respect to m adds a $2\lambda m$ term, which ends up as $+\lambda$ in the denominator.

Differentiate L with respect to m . The squared-error part gives the OLS terms; the penalty adds $2\lambda m$:

$$\frac{\partial L}{\partial m} = -2 \sum_{i=1}^n (y_i - mx_i - b) x_i + 2\lambda m = 0$$

Substitute $b = \bar{y} - m\bar{x}$ and rearrange, as in the OLS Note. The only difference is the extra λm , which joins the other m terms:

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

In words: the top of the fraction is the same as in OLS, and the bottom has λ added. With $\lambda = 0$ the formula is the OLS slope.

Figure 2 shows the mechanism. With the intercept at its best value for each slope, the loss is a curve in m : the squared error (blue) plus the penalty λm^2 (orange). The penalty is a bowl centred on 0, so adding it pulls the bottom of the sum towards 0. As λ grows from 0 to 100, the bottom slides from 27.83 to 12.93, exactly the value of the formula.

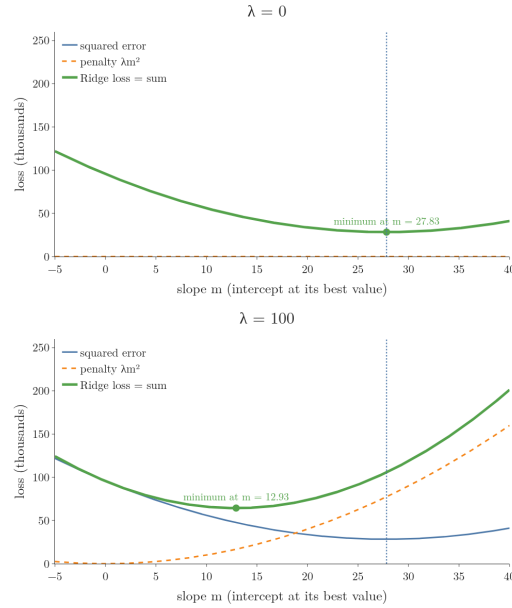


Figure 2: The Ridge loss of the 100-observation example as a function of the slope m , as λ grows from 0 to 100: squared error (blue), penalty λm^2 (dashed orange) and their sum (green). The dotted line marks the OLS slope 27.83; the green minimum slides towards 0.

Picture a seesaw: the top of the fraction is how hard the data pulls the slope up, and λ is a weight added to the other side. The heavier the weight, the less the slope rises.

Extra: Strictly, substituting b turns $\sum(y_i - \bar{y})x_i$ into $\sum(y_i - \bar{y})(x_i - \bar{x})$ and $\sum(x_i - \bar{x})x_i$ into $\sum(x_i - \bar{x})^2$. The pairs of sums are equal because $\sum(y_i - \bar{y}) = 0$ and $\sum(x_i - \bar{x}) = 0$, the same trick as in the OLS derivation.

2.4 Why the slope shrinks

Key point: A bigger λ makes the denominator bigger, so the slope gets smaller. The slope approaches 0 but never reaches it.

On the 100-observation example from the previous Note, the top of the fraction is 2416.7 and the bottom without λ is 86.85. So

$$m = \frac{2416.7}{86.85 + \lambda}$$

λ	Slope m	Intercept b
0	$2416.7/86.85 = 27.83$	-2.295
10	$2416.7/96.85 = 24.95$	-2.127
100	$2416.7/186.85 = 12.93$	-1.425

These slopes are exactly the ones scikit-learn gave in the previous Note. The intercept is not penalised, yet it moves too, because $b = \bar{y} - m\bar{x}$ depends on m . Figure 3 shows the whole curve.

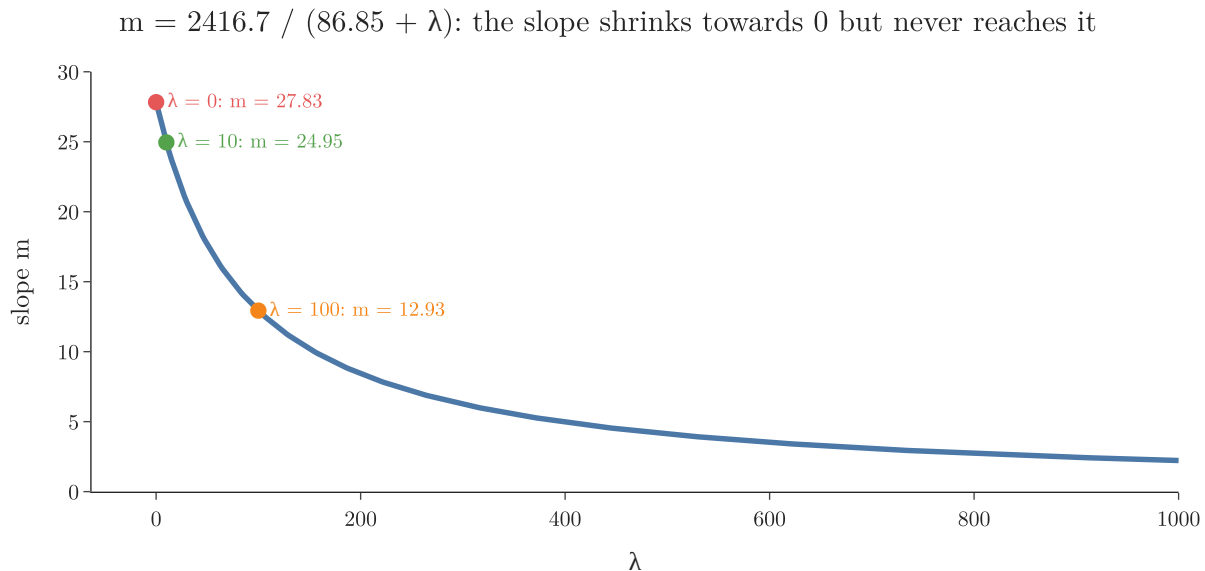


Figure 3: The Ridge slope against λ

However large λ gets, the fraction stays above 0. So Ridge makes coefficients small but never exactly zero. Lasso, two Notes later, behaves differently.

Python: Ridge with one feature, from scratch.

```
class MyRidge:
    def __init__(self, alpha=0.1):
        self.alpha = alpha  # the  $\lambda$  of the formula

    def fit(self, x, y):
        num = np.sum((x - x.mean()) * (y - y.mean()))
        den = np.sum((x - x.mean()) ** 2) + self.alpha
        self.m = num / den
        self.b = y.mean() - self.m * x.mean()
        return self

MyRidge(alpha=10).fit(x, y).m  # 24.955, same as Ridge(alpha=10)
```

3. Many features

Key point: In matrix form the Ridge answer is the normal equation with λI added inside the inverse.

3.1 The loss in matrix form

Key point: The penalty λ times the sum of squared coefficients is written $\lambda w^\top w$.

With many features, the predictions are $\hat{y} = Xw$, where X has a first column of 1s and w holds the intercept and all coefficients (the normal equation Note). The sum of squared coefficients is $w^\top w$, so the Ridge loss is

$$L = (y - Xw)^\top (y - Xw) + \lambda w^\top w$$

3.2 Expanding and differentiating

Key point: The squared-error part gives the same derivative as before; the penalty adds $2\lambda w$.

Expanding the first part exactly as in the normal equation Note gives

$$L = y^\top y - 2w^\top X^\top y + w^\top X^\top X w + \lambda w^\top w$$

Differentiate with respect to w , using the rules from that Note ($w^\top w$ behaves like w^2 , with derivative $2w$):

$$\frac{\partial L}{\partial w} = -2X^\top y + 2X^\top X w + 2\lambda w = 0$$

3.3 Solving for w

Key point: Collect the w terms, factor out w , and multiply by the inverse.

Divide by 2 and move $X^\top y$ to the right:

$$X^\top X w + \lambda w = X^\top y$$

To factor out w , write λw as $\lambda I w$, where I is the **identity matrix** (G-915) (a matrix cannot be added to a number):

$$(X^\top X + \lambda I) w = X^\top y$$

Multiply both sides by the inverse of the bracket:

$$w = (X^\top X + \lambda I)^{-1} X^\top y$$

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

compared with the normal equation, only $+\lambda I$ is new; it adds λ to the diagonal

$$I = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

top-left entry 0: the intercept β_0 is not penalised

Figure 4: The Ridge formula and its identity matrix

Figure 4 shows the result. Compared with the normal equation, only $+\lambda I$ is new: it adds λ to every number on the diagonal of $X^T X$.

3.4 Not penalising the intercept

Key point: The top-left entry of I is set to 0, so the intercept is left out of the penalty.

The first entry of w is the intercept. The intercept only shifts predictions up or down, so it should not be penalised (the previous Note). Setting the top-left entry of I to 0 removes it from the penalty, as in Figure 4.

Python: Ridge with many features, from scratch.

```
class MyRidgeND:
    def __init__(self, alpha=0.1):
        self.alpha = alpha

    def fit(self, X, y):
        X = np.insert(X, 0, 1, axis=1)      # column of 1s
        I = np.identity(X.shape[1])
        I[0, 0] = 0                        # intercept not penalised
        w = np.linalg.inv(X.T @ X + self.alpha * I) @ X.T @ y
        self.intercept_, self.coef_ = w[0], w[1:]
        return self

    def predict(self, X):
        return X @ self.coef_ + self.intercept_
```

On the diabetes data (test size 0.2, random state 4) with alpha 0.1:

	Intercept	Test R^2
MyRidgeND	150.89	0.4693

	Intercept	Test R^2
<code>Ridge(alpha=0.1, solver="cholesky")</code>	150.89	0.4693

The coefficients agree to within 3×10^{-12} , rounding error.

The closed form has the same limit as the normal equation: inverting the matrix costs about the cube of its size in operations, and building $X^T X$ needs the whole dataset in memory. On large data the next Note trains Ridge with gradient descent instead.

Extra: scikit-learn reaches the same answer differently: it centres X and y (subtracts their means), solves for the coefficients without an intercept column, and then computes the intercept from the means. Centring is the matrix version of $b = \bar{y} - m\bar{x}$. The "cholesky" solver (the **Cholesky solver**, G-383) then solves the same equation $(X^T X + \lambda I)w = X^T y$ directly as a **closed-form solution** (G-398), which is why the two rows of the table above agree (scikit-learn docs, **Ridge**).

Extra: A bonus of $+\lambda I$: when columns of X are strongly correlated (**multicollinearity**, G-1273), $X^T X$ can be impossible or unstable to invert. Adding a positive λ to the diagonal always makes it invertible, so Ridge has an answer even when plain OLS does not. In fact this was the main reason Ridge was first introduced (ESL §3.4.1; Hoerl and Kennard 1970). The maths in one line: for any non-zero vector v , $v^T(X^T X + \lambda I)v = \|Xv\|^2 + \lambda\|v\|^2 > 0$, so no v is sent to zero and the matrix has an inverse. The same line covers data with more features than observations, where $X^T X$ never has an inverse: Ridge still returns one answer (ISL §6.2.1).

Figure 5 measures the effect on the diabetes training data. With $\lambda = 0$ the smallest eigenvalue of $X^T X$ is only 0.0073, and the **condition number** (G-441), the largest eigenvalue divided by the smallest, is 48,311: the inverse exists but small rounding errors are blown up. Adding λ lifts the smallest eigenvalue by λ , and at $\lambda = 0.1$ the condition number is already down to about 3,290.

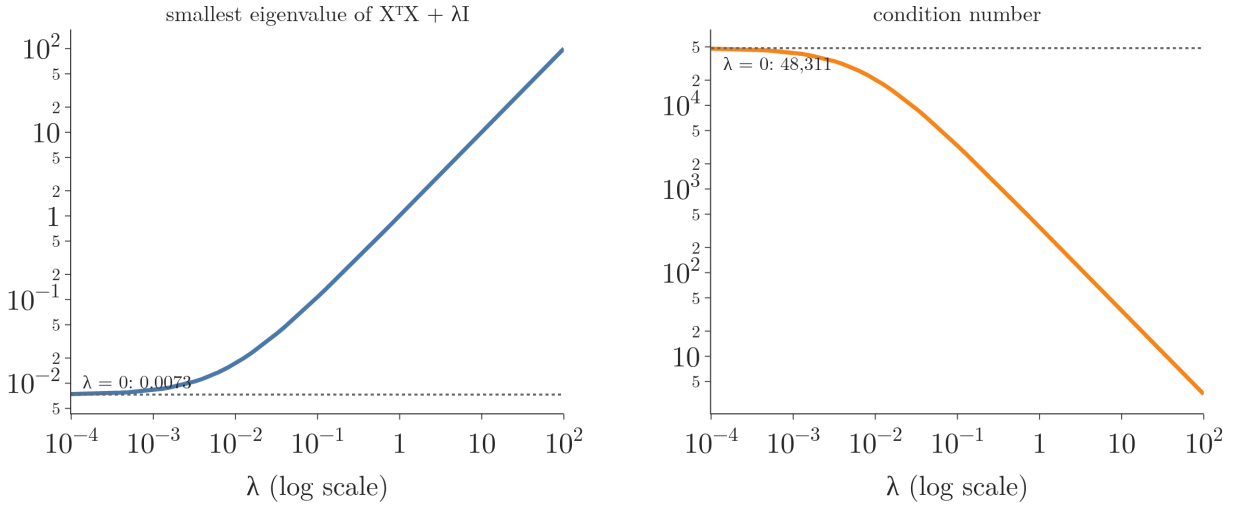


Figure 5: The diabetes training data with its column of 1s. Left: the smallest eigenvalue of $X^T X + \lambda I$ rises with λ from 0.0073. Right: the condition number falls from 48,311 at $\lambda = 0$. Both axes use log scales.

4. Summary

	Linear regression (OLS)	Ridge
Loss	$\sum (y_i - \hat{y}_i)^2$	$\sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$
Slope, one feature	$\frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$	$\frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2 + \lambda}$
Intercept, one feature	$\bar{y} - m\bar{x}$	$\bar{y} - m\bar{x}$
Many features	$(X^T X)^{-1} X^T y$	$(X^T X + \lambda I)^{-1} X^T y$

- Ridge adds λ to the denominator of the slope (one feature) or to the diagonal of $X^T X$ (many features).
- A larger λ gives smaller coefficients, but never exactly 0.
- The intercept is not penalised: the top-left entry of I is 0.
- The from-scratch code matches scikit-learn's **Ridge** exactly.

5. Sources

Built from

- CampusX, “Ridge Regression Part 2 | Mathematical Formulation & Code from scratch | Regularized Linear Models”, YouTube, https://www.youtube.com/watch?v=oDlZBQjk_3A

Other references

- **ESL**: Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Section 3.4.1, p. 64 (which credits Hoerl and Kennard, 1970).
- **Hoerl and Kennard (1970)**: Hoerl, A. E. and Kennard, R. W. “Ridge regression: biased estimation for nonorthogonal problems.” *Technometrics* 12(1), 55–67.
- **ISL**: James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 6.2.1 (Ridge when there are more features than observations).
- **scikit-learn docs**: `sklearn.linear_model.Ridge` (`fit_intercept`, `solver`=“cholesky”), scikit-learn 1.9 documentation; centring step in `sklearn/linear_model/_base.py` (`_preprocess_data`).

6. Key terms

Term	Meaning
Identity matrix	A square matrix with 1s on the diagonal and 0s elsewhere; multiplying by it changes nothing
Closed-form solution	A formula that gives the answer directly, without repeated steps
Cholesky solver	A scikit-learn Ridge solver that solves the closed-form equation directly